Forces on two objects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify an equal-and-opposite interaction pair on different objects.

SUCCESS CRITERIA - I can...

1. Name both objects in an interaction pair.
2. Place the two equal-and-opposite forces on different objects.
3. Explain why equal forces can produce unequal accelerations.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 0.4 kg system accelerates at 0.6 m/s². Predict acceleration for 0.8 kg at the same net force. Explain the effect of greater resistance.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_{A \text{ on } B} = -F_{B \text{ on } A}$.
The two forces act on different objects at the same instant. Their accelerations can differ.

Worked reasoning: Two skaters push apart. Each feels 20 N from the other. If their masses are 40 and 80 kg, accelerations are 0.5 and 0.25 m/s² in opposite directions.

Your turn: Draw separate force diagrams for the horizontal interaction. Explain why equal forces do not imply equal accelerations.



Skater A: 40 kg



Skater B: 80 kg

Marzano check - circle: ① ② ③

03 Equal forces, different results

On a low-friction track, a 1 kg cart and a 4 kg cart push apart. Each interaction force is 12 N. Draw the force on each cart, calculate both acceleration magnitudes, and explain why the forces match but the accelerations do not.



Cart A: 1 kg

Cart B: 4 kg

Marzano check - circle: ① ② ③

04 Identify actual partners

For a book on a table, name the partner of Earth-on-book and the partner of table-on-book. Explain why weight and normal are not a third-law pair.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Horse plus cart

A horse pulls a cart forward while the cart pulls backward on the horse with equal force. Draw a boundary around the two-object system. Add the ground's force on the horse and any resistance. Explain how the combined system can accelerate forward.

Horse

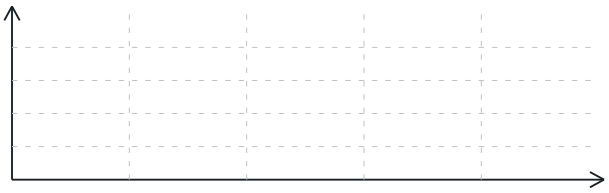
Cart

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

East is positive. Start at 7 m, go to -5 m, then finish at 4 m. Sketch the route; calculate distance and displacement. Distinguish both from final position.
Label the model with the given positions and any prediction.

Route



Position (m)

Marzano check - circle: 1 2 3

07 Exit check

A spacecraft pushes an asteroid that has much greater mass. Draw the interaction force on each object. Compare their forces and the acceleration caused by this interaction. State what you assume about other forces.

Spacecraft

Asteroid

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg cart pushes a 3 kg cart along a frictionless track. An external 10 N force pushes the 2 kg cart to the right. Draw separate force diagrams, find their shared acceleration, and calculate the contact force.



2 kg; applied 10 N right

3 kg

Marzano check - circle: 1 2 3